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AHEAD Online

Object Oriented Programming Using Java

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Objectives

- ✓ Access Specifiers.

Access Control

- ✓ By controlling access, you can **prevent misuse**.
- ✓ **Encapsulation** provides access controls in Java.
- ✓ We can control what parts of a program can access the members of a class.
- ✓ A class creates a blackbox and protect inner working of class from tampering.

Access Specifiers

- ✓ Role of access specifiers is to determine how a member can be accessed.
- ✓ Java's access specifiers are
  - ✓ public
  - ✓ private
  - ✓ protected

Access Specifiers

public

- ✓ When a member of a class is modified by the public specifier, then that member can be accessed by any other code.

private

- ✓ When a member of a class is specified as private, then that member can only be accessed by other members of its class.

Protected

- ✓ accessible within the same package or subclasses in different packages.

Why main( ) has always been preceded by the public specifier?

- ✓ **It is called by code that is outside the program.**
- ✓ Default access specifier is **public.**

Default

- ✓ When no access modifier is specified ,accessible only within the same package

Access Specifiers to data members

Giving access specifiers to data members

private int c; // private access
public int b ;//public access

An access modifier precedes the rest of a member's type specification.

Example Program

```
/* This program demonstrates the difference between public and private. */
class Test {
  int a; // default access
  public int b;  // public access
  private int c; // private access

  // methods to access c
  void setc(int i) {c = i; }
  int getc() { return c; }
}
class AccessTest {
  public static void main(String args[]) {
      Test ob = new Test();
      // These are OK, a and b may be accessed directly
      ob.a = 10; ob.b = 20;
      // This is not OK and will cause an error //
      ob.c = 100; // Error!
      // You must access c through its methods
      ob.setc(100); // OK
      System.out.println("a, b, and c: " + ob.a + " " + ob.b + " " + ob.getc());
  }
}
```

Takeaways

- ✓ Access Specifiers.
- ✓ Types of access specifiers

Reference

[1] Herbert Schildt, *"Java: The Complete Reference, Eleventh Edition", Oracle 2018*